

Hyperjump via the Fourth Spatial Dimension: A Programme Note on Chord Trajectories through the Bulk of a Closed Three-Sphere Universe

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Abstract

Within the Quaternion-Hypersphere Theory of Spacetime [Nopper 2025a], the spatial universe is a closed three-sphere S^3 of geometric radius $R_{\text{geom}}(\tau)$ embedded in a four-dimensional Euclidean space with coordinates $X^A = (\xi, x, y, z)$, often packaged as the quaternion $q = \xi + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. This note examines whether two points of S^3 separated by an angular distance θ can be connected by a trajectory that leaves the hypersurface, traverses the interior of the embedding $\mathbb{R}^4$ as a straight chord, and re-enters at the destination. The chord is strictly shorter than any geodesic arc on S^3 ; in operational terms — for an observer confined to S^3 — the traveller arrives sooner than a light signal travelling along S^3 would, so the trajectory is superluminal in S^3 projection while remaining timelike in the bulk metric. As a candidate dynamical skeleton we propose a separate quaternionic linear sigma model with order-parameter field $\phi^A(X)$ and vacuum expectation value v ; the intended physical brane is the locus near $|X| = R_{\text{geom}}$ where Standard-Model zero modes are localized by the background profile of ϕ^A . The resulting field equations are globally smooth on $\mathbb{R}_\tau \times \mathbb{R}^4$, including at $\phi^A = 0$, and chord traversal becomes either a classical over-the-barrier excitation of the localization field or a quantum-tunnelling process. Eight substantial pieces remain to be supplied — cosmic expansion, coupling to gravity, Standard-Model matter content, the Goldstone-mode mass problem, a fixed value for the coupling λ and vev v , reconciliation with the intrinsic light-speed bound of [Nopper 2025a], a field-theoretic tunnelling-rate calculation, and a UV completion of the (non-renormalizable) quartic interaction. These open subproblems are listed explicitly with first-pass closures. The note is therefore best read as a *programme* rather than a finished theory; it makes no unique precision predictions for laboratory observables beyond those already implied by the underlying braneworld and sigma-model literature, and there is no experimental evidence for macroscopic traversal of extra dimensions.

Keywords: braneworld cosmology; linear sigma model; quaternion-hypersphere spacetime; extra dimensions; superluminal trajectories; Coleman bounce.

1 Introduction

Status and limitations. This document is a *programme note*, not a finished physical theory. It assumes a closed S^3 spatial topology (a premise of [Nopper 2025a], not established by observation), it does not construct a self-consistent 5D gravity + Standard-Model-localization solution, and it does not contain a chronology-protection proof that the bulk-light-cone reformulation in Subproblem 6 is free of closed timelike curves. The “first-pass closures” given for the eight open subproblems are pointers to the appropriate textbook tools, *not* completed calculations for this specific setup. Readers should treat all “in-principle possible” statements as conditional on those three items.

The possibility of connecting widely separated points of the observable universe by a higher-

dimensional shortcut is a recurring theme of speculative physics, from the Einstein–Rosen bridge and traversable wormholes [Morris & Thorne 1988] to the Alcubierre warp metric [Alcubierre 1994]. These constructions invariably require either non-trivial topology of the spacetime manifold or sources that violate the standard energy conditions of general relativity. In contrast, the Quaternion-Hypersphere Theory of Spacetime [Nopper 2025a] already comes equipped with a natural higher-dimensional embedding: the spatial universe is taken to be a closed three-sphere S^3 sitting inside an auxiliary four-dimensional Euclidean space $\mathbb{R}^4$. The present note explores the consequences of taking that embedding seriously as a physical arena rather than as a purely auxiliary mathematical construct.

The paper is organized as follows. Section 2 establishes the elementary geometric statement that a straight chord through $\mathbb{R}^4$ is strictly shorter than any geodesic arc on S^3 between the same two points. Section 3 introduces a quaternionic linear sigma model as a candidate dynamical skeleton on the bulk and identifies the parametric scales it generates. Section 4 enumerates eight open subproblems that must be closed before the proposal can be considered a candidate physical theory, with first-pass closures pointing to the appropriate textbook tools. Section 5 outlines a tiered experimental programme and explicit falsification criteria. Section 6 comments on the relationship to the interval-number framework of [Nopper 2025b]. Section 7 summarizes the status of the proposal.

2 The Geometric Argument

A point on the spatial hypersphere satisfies $|X| = R_{\text{geom}}$. (Note: that the spatial universe is closed with S^3 topology is a *premise* taken from [Nopper 2025a]; current cosmological data are consistent with this premise but do not establish it — see §Experimental Programme.) Two points $X_1, X_2 \in S^3_{R_{\text{geom}}}$ separated by angular distance θ admit two path types: a geodesic *arc* of length $R_{\text{geom}}\theta$ that stays on S^3 , and a straight-line *chord* through the embedding $\mathbb{R}^4$ of length $2R_{\text{geom}}\sin(\theta/2)$. The two-dimensional analogue ($S^1 \subset \mathbb{R}^2$) is shown in Figure 1.

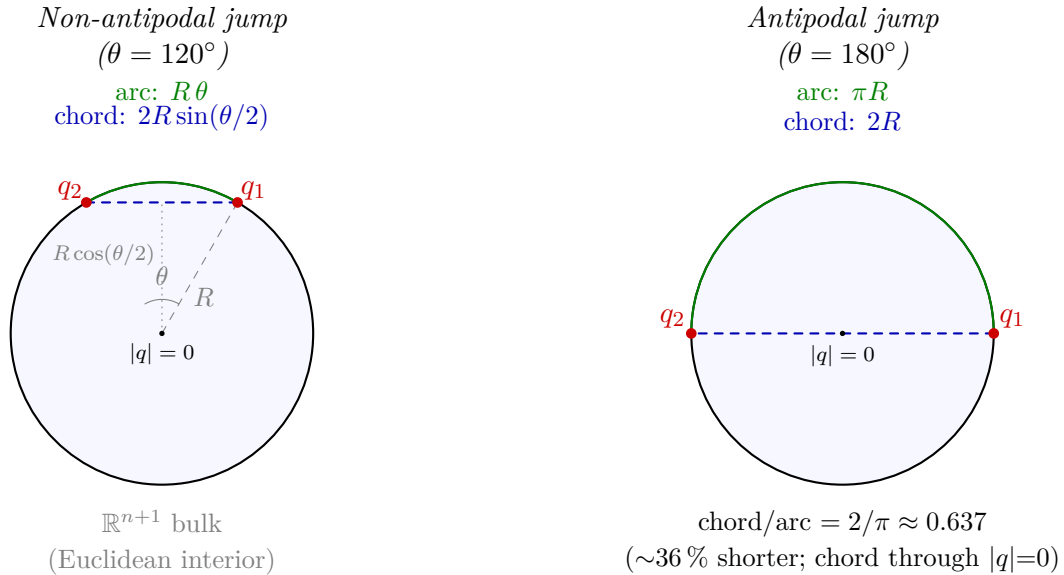


Figure 1: Chord through $\mathbb{R}^{n+1}$ versus geodesic arc on S^n (illustrated for $n = 1$). The hyperjump trajectory (dashed blue) leaves S^n , traverses $\mathbb{R}^{n+1}$, and re-enters at q_2 .

Path	Parameterization	Length
Geodesic on S^3 (normal travel)	$\text{slerp}(X_1, X_2, t)$, with $ X(t) = R_{\text{geom}}$	$R_{\text{geom}} \theta$
Chord through $\mathbb{R}^4$ bulk (hyperjump)	$(1-t)X_1 + tX_2$, with $ X(t) \leq R_{\text{geom}}$	$2R_{\text{geom}} \sin(\theta/2)$

Since $\sin(\theta/2)/(\theta/2) \leq 1$ for all $\theta \in (0, \pi]$, the chord is never longer than the arc. The maximum shortcut occurs at the antipodal point ($\theta = \pi$): chord $= 2R_{\text{geom}}$ versus arc $= \pi R_{\text{geom}}$ — a reduction of approximately 36%. At bulk speed $u \leq c$ the crossing time is $\Delta\tau = 2R_{\text{geom}} \sin(\theta/2)/u$, strictly finite and timelike with respect to the bulk metric.

It must be acknowledged that, from the perspective of an S^3 -confined observer, the chord traverser arrives sooner than a light signal travelling along S^3 between the same endpoints. The effective S^3 -projected speed for an antipodal chord at bulk speed c is $\pi c/2 \approx 1.57c$. The trajectory is therefore operationally superluminal in S^3 projection. This is consistent with the bulk light-cone but in tension with the FTL prohibition in [Nopper 2025a, §Light-Speed Bound], which is formulated *intrinsically* on S^3 . The resolution adopted here — extending the causal-cone definition to the bulk Minkowski metric — is given as the first-pass closure of subproblem 6 below.

For an antipodal jump the chord passes through $|X| = 0$ at its midpoint; for non-antipodal jumps it passes through a minimum norm $R_{\text{geom}} \cos(\theta/2) > 0$. The bulk origin $|X| = 0$ is a point of the auxiliary embedding $\mathbb{R}^4$ at the present cosmic time τ ; it is *not* the FLRW initial singularity at $\tau = 0$, although both share the property that the geometric radius vanishes.

3 A Candidate Dynamical Skeleton: Quaternionic Linear Sigma Model

The SpacetimeTheory designates $\mathbb{R}^4$ as an *auxiliary* embedding space and defines matter only on S^3 [Nopper 2025a, §Foundations]. To make the hyperjump a physical process, the embedding coordinate X^A must be supplemented by independent dynamics: an action principle for fields living on the bulk that admits both S^3 -localized ground states and chord-like excited trajectories.

We propose, as the simplest candidate skeleton, a Lorentz-invariant linear sigma model with global $O(4)$ symmetry on the bulk $\mathcal{B} = \mathbb{R}_\tau \times \mathbb{R}^4$. The dynamical field is a real 4-component order parameter $\phi^A(X)$ ($A = 1, \dots, 4$), not the embedding coordinate X^A itself:

$$\mathcal{L}_\phi = -\frac{1}{2} \eta^{MN} \partial_M \phi^A \partial_N \phi^A - \lambda (\phi^A \phi^A - v^2)^2 \quad (1)$$

with five-dimensional Minkowski signature $\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1)$ in natural units ($\hbar = c = 1$) and implicit sums over M, N and A . Structurally this is the Higgs sector with the field-space vacuum manifold $\{\phi^A \phi^A = v^2\} \cong S^3$. The physical spatial hypersphere remains the geometric locus $|X| = R_{\text{geom}}$; identifying that locus with the low-energy brane requires a background profile $\phi_0^A(r)$, where $r = |X| - R_{\text{geom}}$, that localizes Standard-Model zero modes near $r = 0$. We emphasize that such a radial domain-wall profile at a specific finite radius $|X| = R_{\text{geom}}$ does *not* form spontaneously in flat $\mathbb{R}^4$ from the potential $\lambda(|\phi|^2 - v^2)^2$ alone — the natural codimension-four defect of an $O(4)$ -symmetric model is a global-monopole hedgehog $\phi^A \propto \hat{X}^A$ centred at $|X| = 0$, not a thin shell at finite radius. Pinning the brane at $|X| = R_{\text{geom}}(\tau)$ requires the additional dynamical input of Subproblems 1 and 2 (cosmic-expansion sector and coupling to gravity); $\phi_0^A(r)$ should therefore be read as the ansatz one wishes a self-consistent solution to produce, not as an established solution of the present action.

Conventions and dimensions. Throughout this note we use natural units $\hbar = c = 1$ unless otherwise stated. In five spacetime dimensions a canonically normalized real scalar ϕ^A has mass

dimension $3/2$; the vev v has mass dimension $3/2$; and λ has mass dimension -1 (confirming non-renormalizability — see Subproblem 8). The geometric radius R_{geom} has dimension of length and is not the same object as v . A complete gravity/localization solution must determine how v , λ , and the background profile $\phi_0^A(r)$ set the brane thickness and the observed geometric radius.

The skeleton does three things:

Feature	Mechanism
Brane localization becomes dynamical	Matter confinement is controlled by $\phi_0^A(r)$ rather than imposed as the kinematic constraint $ X = R_{\text{geom}}$
Equations of motion are globally smooth	$\square\phi^A = 4\lambda(\phi ^2 - v^2)\phi^A$ is well-posed everywhere on the bulk, including at $\phi^A = 0$
The central field configuration is non-singular	$V(0) = \lambda v^4$ is finite; leaving the brane encounters a finite barrier, not a divergence

Excitations with energy density above the central barrier $\rho_\star = \lambda v^4$ can in principle push localized states into the bulk along chord trajectories; excitations below the barrier tunnel quantum-mechanically. The semiclassical estimate must, however, be adapted to the symmetry of the potential: $V(\phi) = \lambda(\phi^A\phi_A - v^2)^2$ has two *degenerate* minima on the vacuum manifold $|\phi| = v$, so the standard Coleman false-vacuum bounce [Coleman 1977] does not apply — there is no finite-action $O(d)$ -symmetric instanton that starts at $|\phi| = v$, passes through $|\phi| = 0$, and returns. The correct saddle is instead the planar domain wall (kink) of surface tension

$$\sigma = \frac{4\sqrt{2}}{3} \sqrt{\lambda} v^3, \quad (2)$$

and the Euclidean action for a cargo of transverse 3-area A_c traversing a chord of length $\ell_{\text{chord}} = 2R_{\text{geom}} \sin(\theta/2)$ is dominated by the worldvolume of the wall-tube wrapping that cargo,

$$B_{\text{phys}} \sim \sigma (P_c \ell_{\text{chord}} + 2 A_c), \quad (3)$$

with P_c the 2-perimeter of the cargo transverse to the chord. Once gravity is included, the appropriate extension of (3) is a Coleman–De Luccia-type calculation [Coleman & De Luccia 1980] for the wall worldvolume on the curved bulk, not for a Coleman bounce. The parameters λ and v then control the *brane thickness* $\ell_\perp \sim 1/(\sqrt{\lambda}v)$, the *barrier energy density* $\rho_\star = \lambda v^4$, the *wall tension* (2), and via (3) the parametric *action cost* of transporting an object through the bulk.

This much is a clean dynamical skeleton. It is **not yet a complete physical theory**.

4 Open Subproblems

The skeleton above leaves eight substantial pieces unsupplied. Each is an explicit open problem; the model becomes a candidate physical theory only when all are addressed. Each item is given a textbook-level *first-pass closure* inline — a starting point, not a final solution. The depth of the residual work varies considerably: items 1–5 reduce to standard exercises in scalar-field cosmology, braneworld gravity, brane localization, gauge symmetry breaking, and dimensional analysis; items 6–7 require dedicated technical follow-ups (causal-structure reformulation, full bounce computation); item 8 is a deferral to a UV-completion programme that lies outside the scope of any single note.

1. **Cosmic expansion.** [Nopper 2025a] has $R_{\text{geom}} = R_{\text{geom}}(\tau)$ (FLRW). The static brane location in the skeleton is incompatible. *First-pass closure*: introduce an additional collective coordinate

$\Phi(\tau)$ controlling the equilibrium brane radius and a potential $U(\Phi)$ with a kinetic term $-\frac{1}{2}(\partial\Phi)^2$. (We emphasise: Φ is an effective field controlling the brane location in the embedding $\mathbb{R}^4$; it is *not* the FLRW scale factor itself, which must instead emerge from the induced 4D metric of Subproblem 2.) The homogeneous mode then satisfies $\ddot{\Phi} + 3H\dot{\Phi} + U'(\Phi) = 0$, the standard scalar-field-cosmology equation [Linde 1990]; for slow-roll choices of U this reproduces de Sitter expansion. Full closure requires coupling to 5D gravity (Subproblem 2) so that H , the relation between Φ and the sigma-model profile, and the resulting FLRW scale factor are all solutions rather than postulates.

2. **Coupling to gravity.** The bulk is treated as flat. To recover the FLRW geometry as a *solution* rather than a postulate, the action must be coupled to 5D General Relativity,

$$S = \int d^5X \sqrt{-g} \left[\frac{\mathcal{R}^{(5)}}{2\kappa_5^2} + \mathcal{L}_\phi \right], \quad (4)$$

in the spirit of Randall–Sundrum-type braneworlds. *First-pass closure:* in the thin-brane limit $\ell_\perp \rightarrow 0$, the energy-momentum tensor of the localized $\phi_0^A(r)$ background degenerates to a delta-function shell source near $|X| = R_{\text{geom}}$ and the 5D Einstein equations reduce to Israel junction conditions [Israel 1966]; the induced 4D geometry is then standard FLRW with the brane tension $\sigma = \int dr V(\phi_0)$ playing the role of an effective cosmological constant. A treatment with finite $\ell_\perp$ requires solving the coupled (g_{MN}, ϕ^A) system numerically; with the warped ansatz $ds^2 = e^{2A(r)} \eta_{\mu\nu} dx^\mu dx^\nu + dr^2$ the system admits an explicit DeWolfe–Freedman–Gubser–Karch first-order flow $\phi'(r) = W'(\phi)$, $A'(r) = -W(\phi)/3$ with induced potential $V(\phi) = \frac{1}{2}W'(\phi)^2 - \frac{2}{3}W(\phi)^2$ [DeWolfe et al. 2000]. Choosing $W(\phi) = \phi - \phi^3/(3v^2)$ gives the closed-form domain-wall background $\phi_*(r) = v \tanh(r/v)$, $A_*(r) = -\frac{2}{9} \ln \cosh(r/v) - \frac{1}{18} \tanh^2(r/v)$, which we reproduce numerically with a physics-informed neural network in the companion code (`numerics/pinn_coupled.py`). This is an explicit self-consistent 5D background of the type the hyperjump skeleton requires.

3. **Standard-Model matter on the brane.** Real matter is not a quaternion sigma model. *First-pass closure:* introduce a 5D Dirac fermion Ψ with a Yukawa mass profile $m_\Psi(r) = h f[\phi_0(r)]$ chosen so that $m_\Psi(r)$ changes sign across the brane at $r = |X| - R_{\text{geom}} = 0$ [Rubakov–Shaposhnikov 1983]. The transverse Dirac equation $[i\gamma^\perp \partial_r - m_\Psi(r)]\Psi = 0$ then admits a normalizable zero mode with profile $\Psi_0(r) \propto \exp[-\int^r m_\Psi(r') dr']$ peaked on S^3 and exponentially suppressed in the bulk; this is the 4D Standard-Model fermion. Gauge bosons can be localized by analogous dilaton-type couplings [Dvali–Shifman 1997]. Recovering the full SM spectrum (chirality, generations, Higgs) within this framework is non-trivial but follows established braneworld phenomenology.
4. **Goldstone-mode mass problem.** The breaking $O(4) \rightarrow O(3)$ in field space produces three exactly massless scalars not seen in nature. *First-pass closure:* gauge the connected component $SO(4)$ by introducing gauge fields A_M^a ($a = 1, \dots, 6$, since $\dim SO(4) = 6$) and replacing the partial derivative by the covariant derivative,

$$\partial_M \phi^A \longrightarrow D_M \phi^A = \partial_M \phi^A - ig A_M^a (T^a)^A_B \phi^B, \quad (5)$$

plus a Yang–Mills kinetic term $-\frac{1}{4}F_{MN}^a F^{aMN}$. After symmetry breaking, the three broken generators give three massive vector bosons with masses $m_A \sim gv$; the three Goldstones become their longitudinal components via the Higgs mechanism. The remaining $SO(3)$ gauge bosons stay massless and would need to be either confined or identified with a known sector — itself a non-trivial phenomenological constraint.

5. **Determination of λ and v .** The coupling and vev are unconstrained, so the model is not predictive. *First-pass closure:* in the present setup the transverse direction (radial in the embedding $\mathbb{R}^4$, across the S^3 brane) is *non-compact*, so $\ell_\perp \sim 1/(\sqrt{\lambda}v)$ is a brane-thickness

/ warping length, not a Kaluza–Klein compactification radius. Current laboratory bounds on deviations from the Newtonian inverse-square law constrain a new transverse length scale to $\ell_\perp \lesssim 52 \mu\text{m}$ at 95 % C.L. [Lee et al. 2020], superseding the earlier $\sim 44 \mu\text{m}$ bound reviewed in [Adelberger et al. 2003]; this translates into $\sqrt{\lambda} v \gtrsim 1/\ell_{\text{exp}} \sim 4 \text{ meV}$ at the present level of precision. This constrains only the product $\sqrt{\lambda} v$; pinning unique values requires an independent observable (e.g. a continuum graviton signature in a non-compact transverse direction, a tunnelling rate, a radial-mode mass, or a cosmological constraint on Φ). The precise translation of the laboratory bound into a constraint on $\sqrt{\lambda} v$ depends on the unsolved gravity sector (Subproblem 2) and is therefore itself only first-pass.

6. **Operational FTL versus [Nopper 2025a]’s prohibition.** Chord traversal is timelike in the bulk but superluminal in S^3 projection. *First-pass closure:* adopt the bulk Minkowski light-cone of $\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1)$ (natural units) as the *fundamental* causal structure, and reinterpret [Nopper 2025a]’s intrinsic light-speed bound as an emergent statement about S^3 -confined matter — strictly valid only for trajectories that remain on the brane. Chord trajectories then satisfy the bulk light-cone. The standard “FTL implies CTCs by Lorentz-boosting” argument does *not* directly apply here, because the FLRW embedding picks out a preferred cosmic-time slicing and brane Lorentz boosts are not a true symmetry of the full theory. *However*, this observation alone does not constitute a chronology-protection proof: brane-confined observers in different regions of S^3 enjoy *approximate* local Lorentz invariance, and one must verify explicitly that no concatenation of (i) brane-confined sub-light trajectories and (ii) allowed bulk chords can close into a causal loop with respect to the bulk light-cone. We do not supply that verification here; it is an explicit open subproblem of this note. The cost is conceptual: the “intrinsic” formulation of light-speed in [Nopper 2025a] must be downgraded to an effective brane-projected bound, and Lorentz invariance of the brane physics becomes approximate rather than exact (constrained by precision tests of vacuum Lorentz invariance — see Tier 1 below). An alternative — adding a chronology-protection term that suppresses access to $\phi^A \approx 0$ — preserves the intrinsic formulation but reintroduces an arbitrary suppression scale.

7. **Tunnelling rate.** A field-theoretic semiclassical calculation should replace the 1D-WKB heuristic. *First-pass closure:* the symmetric double-well $V(\phi) = \lambda(\phi^A \phi^A - v^2)^2$ has two *degenerate* vacua on $|\phi| = v$, so a Coleman-type $O(d)$ -symmetric bounce [Coleman 1977] interpolating between $|\phi| = v$ at infinity and $|\phi| = 0$ at the centre does not exist as a finite-action saddle — the relevant process is not false-vacuum decay. The appropriate saddle is the planar kink with surface tension $\sigma = \frac{4\sqrt{2}}{3} \sqrt{\lambda} v^3$ (Eq. 2); the Euclidean action for a cargo traversing a chord of length ℓ_{chord} is dominated by the wall-tube worldvolume,

$$B_{\text{phys}} \sim \sigma(P_c \ell_{\text{chord}} + 2A_c),$$

giving $\Gamma/\mathcal{V}_5 \sim \mu^5 e^{-B_{\text{phys}}/\hbar}$. Substituting the laboratory-saturating value $\sqrt{\lambda} v \sim 1/\ell_\perp \sim 4 \text{ meV}$ from Subproblem 5 and any macroscopic cargo (e.g. a 1 m-radius object across a 1 ly chord) gives $B_{\text{phys}} \gtrsim 10^{30}$, so unaided hyperjumps are exponentially suppressed beyond microscopic probes and would require an external coherent source to occur on macroscopic timescales. Once gravity is included, the appropriate extension is a Coleman–De Luccia-type computation [Coleman & De Luccia 1980] of the wall worldvolume on the curved bulk; this fixes the exponent’s $O(1)$ coefficient and the prefactor. A companion numerical implementation (`numerics/bounce.py` in this repository) verifies σ analytically and numerically and tabulates B_{phys} for sample cargos.

8. **UV completion.** The quartic interaction $\lambda(\phi^A \phi^A - v^2)^2$ is non-renormalizable in five spacetime dimensions; the Lagrangian is therefore at best an effective theory below some cutoff Λ . *First-pass framing:* treat the action as a Wilsonian EFT valid for energies $E \ll \Lambda \lesssim M_{\text{Pl}}^{(5)}$

(the 5D Planck mass after Subproblem 2 is implemented). Within that range the model is predictive in the EFT sense — operators of dimension > 5 are suppressed by powers of E/Λ — which suffices for any prediction at sub-Planckian scales, including the chord-traversal energetics. A genuine UV completion (asymptotic safety, embedding in a higher-symmetry or string-theoretic theory) lies beyond the scope of this note and is not supplied here.

A consistent closure of items 1–2 alone would already promote the proposal from a kinematic skeleton to a candidate dynamical theory; items 3–5 and 7 would make it phenomenologically meaningful; item 6 reframes the causal structure of [Nopper 2025a]; item 8 is a deferral to a UV-completion programme.

5 Experimental Programme

The model’s predictions split cleanly into two regimes: properties of the *underlying sigma-model and brane structure*, which are testable with existing or near-term experiments, and properties of *macroscopic chord traversal itself*, which are not testable with any foreseeable technology.

5.1 Tier 1 — Tests parasitic on running or near-term experiments

Observable	What it constrains	Status
Sub-mm gravity (Eöt-Wash and successors)	Brane thickness $\ell_\perp$ constrains $\sqrt{\lambda}v$ via Subproblem 5	Currently $\ell_\perp \lesssim 52\,\mu\text{m}$ (95 % C.L.) [Lee et al. 2020], superseding $\sim 44\,\mu\text{m}$ [Adelberger et al. 2003]; next-generation aims at the micron scale
Fifth-force / new long-range vector searches	Residual unbroken $SO(3)$ gauge bosons (Subproblem 4) must be confined or below sensitivity	Tight bounds on any massless vector mediator
Vacuum Lorentz-invariance / preferred-frame tests	Approximate brane Lorentz invariance required by Subproblem 6	Existing bounds require any bulk-frame effects to be strongly suppressed
CMB scalar/tensor ratio (Planck, LiteBIRD, CMB-S4)	Shape of the inflaton potential $U(\Phi)$ in Subproblem 1	Tightening r disfavors specific $U(\Phi)$
Spatial-curvature and topology tests (CMB matched circles)	The model presupposes a closed three-sphere universe of finite radius R_{geom}	Data consistent with flat or slightly closed; no positive detection

5.2 Tier 2 — Dedicated or next-generation facilities

Observable	What it constrains
KK-like resonances and missing-energy events at HL-LHC / FCC-hh	Bulk-propagating excitations of ϕ^A ; matter briefly entering the bulk
Precision Higgs couplings	Mixing between the SM Higgs and the radial (massive) mode of ϕ^A
Dark-energy equation of state $w(z)$	Trajectory of $\Phi(\tau)$ on $U(\Phi)$

5.3 Tier 3 — The hyperjump itself

Given the exponential suppression in Subproblem 7, no terrestrial experiment can plausibly produce a macroscopic chord traversal. Conceptual milestones, in increasing order of difficulty:

1. **Microscopic tunnelling event.** Observation of a single quantum appearing at an unexpected location with a rate consistent with the bounce calculation, after excluding all conventional channels. Astronomically rare for any currently allowed (λ, v, ϕ_0) background.
2. **Coherent-source proof of principle.** A configuration raising ϕ^A above the central barrier in a small region. The required energy density $\rho_\star = \lambda v^4$ is bounded below indirectly through Subproblem 5's lower bound on $\sqrt{\lambda}v$, and for any macroscopic cargo is extreme by any laboratory standard — qualitatively comparable to the absolute energy-scale problem in warp/wormhole proposals, though here with positive sign.
3. **Macroscopic chord traversal.** Direct observation of an object appearing near its antipode in less than light-travel time on S^3 . Not forbidden by the framework, but on no plausible engineering horizon.

5.4 Falsification criteria

The proposal is *falsified* (in the strong Popperian sense) if any of the following obtains:

- Cosmological curvature and topology constraints drive the allowed parameter interval entirely outside the closed three-sphere geometry required by [Nopper 2025a];
- Sub-mm gravity, fifth-force, and localization constraints leave no overlapping parameter region for $\ell_\perp$, λ , v , and the Standard-Model zero-mode profiles;
- Precision tests of Lorentz invariance in vacuum exclude the bulk light-cone reformulation of Subproblem 6 at the level required for the chord trajectories to remain consistent with observed brane physics;
- A graviton spectrum inconsistent with a single non-compact transverse dimension — for example, a sharp Kaluza–Klein tower with discrete mass spacing characteristic of a *compact* extra dimension, rather than the localized zero mode plus continuum expected for a non-compact transverse direction in the present setup — is established at any future collider or astrophysical observation.

5.5 Honest summary

The *theory* is empirically engageable through existing programmes; Tier 1 can strongly constrain it and could potentially falsify specific parameter regions. The *device* — hyperjump propulsion — is not on any plausible engineering horizon and would require physics or technology beyond what is currently available.

5.6 In-principle possibility

Subject to the obvious disclaimers, the framework contains no in-principle prohibition against constructing a hyperjump device. What it contains are quantitative obstacles: depositing energy density $\rho_\star = \lambda v^4$ coherently in the cargo volume, maintaining quantum coherence of the bulk excitation across the chord, and either accepting the exponentially small spontaneous tunnelling rate or engineering a resonant/stimulated enhancement. Each is severe; none is forbidden by the present skeleton.

In this respect the proposal sits in the same epistemic category as traversable wormholes [Morris & Thorne 1988] and the Alcubierre warp metric [Alcubierre 1994]: physically permitted within a self-consistent theoretical framework, but with engineering requirements far beyond present extrapolation. One distinguishing structural feature is worth noting: the chord traversal **does not require violation of the energy conditions of general relativity** — it requires positive-energy excitations of the sigma-model field above a finite barrier $V(0) = \lambda v^4$, whereas

wormhole and Alcubierre constructions require negative energy density (exotic matter). Whether this counts as a *practical* advantage is debatable, since the absolute energy-density scale required here is itself extreme; what is unambiguous is that the *qualitative* obstacle is different.

The honest classification is therefore: *in-principle possible, technologically remote, and structurally distinct from negative-energy-density proposals.*

6 Note on the Interval-Number Framework

Earlier formulations of this proposal invoked the interval-number algebra of [Nopper 2025b] to regularize formal expressions of the type $\rho_{\text{bulk}} \cdot V_{\text{bulk}} \sim |X|^{-3} \cdot |X|^3 \rightarrow 0 \cdot \infty$ that arise if one extends FLRW-style densities into the bulk via the geometric radius $|X|$. Within the sigma-model skeleton above, no such densities appear: the action is governed by the Lagrangian $\mathcal{L}_\phi$, which is finite everywhere. The interval-number framework therefore plays no role in the present formulation. It remains an independent, self-consistent algebraic system [Nopper 2025b] and may still be useful for other extensions of the theory in which genuinely indeterminate forms appear.

7 Summary

Framework	Role
Quaternion-Hypersphere Theory [Nopper 2025a]	Identifies $\mathbb{R}^4$ as the embedding space and supplies the chord-path geometry of the closed S^3 universe
Quaternionic linear sigma model (this note)	Provides a candidate bulk Lagrangian for an order-parameter field ϕ^A whose background profile localizes matter near S^3 and whose excited states can enter the bulk

The model is **geometrically coherent and singularity-free**, but **not yet dynamically complete**: the eight open subproblems listed above must be closed before it can be considered a candidate physical theory. Each admits a textbook-level first-pass closure sketched inline (with item 8 framed as a Wilsonian-EFT deferral), but full closure of any one item is itself a non-trivial research programme. It introduces, beyond [Nopper 2025a], the $O(4)$ -symmetric quartic potential $V(\phi) = \lambda(\phi^A \phi^A - v^2)^2$ with coupling λ and vev v . These parameters have clean physical interpretations (barrier energy density $\rho_\star = \lambda v^4$, brane thickness $\ell_\perp \sim 1/(\sqrt{\lambda} v)$, transit-energy cost $\rho_\star V_{\text{cargo}} \ell_{\text{chord}}$) but no fixed numerical values at present. The model makes no unique precision predictions yet, and there is no experimental evidence for macroscopic bulk traversal.

References

Foundations of this work

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Extra dimensions and braneworld models

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Sigma models, symmetry breaking, brane localization, tunnelling

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Background (Wikipedia)

- [N-sphere](#)

- [Quaternion](#)
- [Linear sigma model](#)
- [Brane cosmology](#)
- [Kaluza–Klein theory](#)
- [Faster-than-light](#)
- [Geodesic](#)

Inspirational background (science fiction)

The motivating question of this note — whether a higher-dimensional shortcut between distant points of our spatial universe could be physically realisable — is a recurring motif in twentieth-century science fiction. The works listed below served only as *inspiration* for posing the question in the first place; they form no part of the physics or mathematics presented above, and no claim is made that the corresponding fictional devices realise the proposed mechanism.

- [Asimov 1951] Asimov, I. *Foundation*. Gnome Press, 1951. — hyperspatial jumps.
- [Herbert 1965] Herbert, F. *Dune*. Chilton Books, 1965. — foldspace navigation.
- [Roddenberry 1966] Roddenberry, G. *Star Trek* (TV series). NBC, 1966. — warp drive through subspace.
- [Lucas 1977] Lucas, G. *Star Wars: Episode IV – A New Hope*. 20th Century Fox, 1977. — hyperspace.
- [Banks 1987] Banks, I. M. *Consider Phlebas*. Macmillan, 1987. — discrete jump drives.